

Spintronic Systems for the Continuum of Analog and Digital AI Acceleration

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As the computational demands of artificial intelligence (AI) continue to scale at an unprecedented pace, various computing architectures, from digital to analog, are being explored to develop AI accelerators. The recent surge of large language models has created a pressing need for efficient memory-centric computing systems. As spintronics has evolved as the underlying enabling technology for data storage and non-volatile memory for several decades, leveraging spintronics for memory-centric AI computation offers exciting opportunities. In this paper, we present a comprehensive discussion of spintronic computing systems, demonstrating the unique capabilities of spintronics to enable a range of core functionalities across analog and digital computing systems.

Index Terms—Spintronics, AI, Computing, Accelerator

I. INTRODUCTION

THE rapid progress of AI is fueled by the combination of advances in algorithmic innovation (such as deep neural networks) and the proliferation of high-performing computing and storage. However, as AI computation continues to grow, the separation of processing and memory units in the standard von Neumann architecture creates a severe bottleneck when processing data-intensive workloads. At present, addressing the von Neumann memory bottleneck is one of the most important topics in the design of AI hardware accelerators, driving growing interest in various emerging memory-centric computing architectures. In this paper, we will show that spintronic devices and circuits can be designed to realize various critical functionalities for in-/near-memory accelerators. The rich physics in nano-magnetism provides immense opportunities for compute-memory fusion and analog-digital continuum, as well as efficient neuro-synaptic emulations for neuromorphic computing. We demonstrate that hardware-software co-design across the stack from devices/circuits to architectures and algorithms is crucial for achieving optimal processing efficiency and maintaining functional accuracy.

II. SPINTRONIC MEMORY TECHNOLOGIES

Minimizing data movement in AI computation requires large on-chip memory to accommodate various data structures, including activations, weights, and intermediate quantities such as partial sums. As a result, in today's silicon, 50-90% of the total chip area of modern processors and AI accelerators is consumed by memory and memory-related I/O (input/output) [1], [2]. The standard on-chip memory, static random-access memory (SRAM), is based on Complementary-Metal-Oxide-Semiconductor (CMOS) technology and requires 6-8 transistors to construct a memory cell. Importantly, SRAM scaling has stalled at advanced transistor nodes. To achieve improved on-chip density, pursuing area-efficient memory has been a major driving force in the exploration of unconventional and beyond-CMOS device technologies. As summarized in Table I, various emerging non-volatile memory technologies have demonstrated immense potential with the capability of 1-2 transistors per cell, including resistive RAM (ReRAM), phase

TABLE I
COMPARISON OF EMERGING NON-VOLATILE MEMORY WITH CMOS SRAM. (*)MRAM WITH MULTI-DOMAIN AND MULTI-GRANULAR STRUCTURE MAY ACHIEVE 2 BITS PER CELL.

Technology	ReRAM	PCM	MRAM	SRAM
Bits/Cell	1-5	1-3	1 or 2*	1
Cell Density	$16F^2$	$16F^2$	$30-80F^2$	$>150F^2$
Write Energy	2nJ	6nJ	1 nJ	$< 0.1\text{nJ}$
Write Latency	100 ns	>150 ns	2-10 ns	< 1 ns
Write Endurance	10^6	10^7	10^{15}	unlimited

change memory (PCM), and spintronic-based magnetic RAM (MRAM). MRAM offers balanced read/write latencies, the highest write endurance, and the lowest variability among the top NVM candidates. As discussed in the following sections (Fig. 1), we will show that the unique advantages of MRAM enable novel pathways for designing AI hardware.

III. VARIOUS ANALOG AND DIGITAL AI ACCELERATORS

Digital Near-Memory Computing Spin-transfer-torque (STT) MRAM provides the most mature high-performing on-chip non-volatile memory technology supported by major foundries. The high write endurance and near-nanosecond access speed open up exciting opportunities for deployment in digital AI accelerators. Various architectural innovations aim to alleviate the memory bottleneck by provisioning on-chip memory near the processing units. MRAM with a 2-4 density advantage compared to SRAM becomes a strong competitor to implement scratchpad memory with structured and deterministic access patterns and simplified memory hierarchy. Importantly, the write error rate and write energy/latency are highly correlated for STT-MRAM, opening up an interesting design space exploration with efficiency-accuracy trade-off. Device-to-system co-design will include customized memory cell design (both the magnetic tunnel junction and the access transistor), thermal stability and retention configuration, and setting the duration and magnitude of memory write pulses. Both training and inference accelerators can leverage the error resiliency of AI workloads to deliver substantial gains in area efficiency and reduced energy consumption [3], [4].

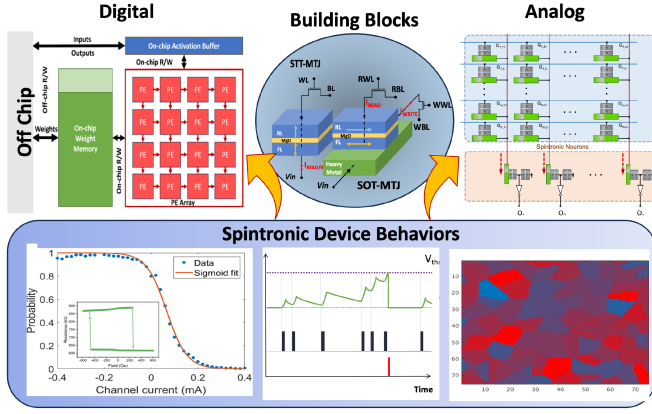


Fig. 1. Spintronic systems for digital and analog AI hardware

Digital In-Memory Computation While near-memory hardware effectively mitigated the von Neumann memory bottleneck, memory and compute units are still physically separate in processing near-memory systems. Compute in-memory with fused memory and compute units further reduces the data movement bottleneck. Both digital and analog computing in-memory have been developed. Digital computation minimizes circuit noises and non-idealities and realizes the underlying Boolean arithmetics within the memory arrays. For example, spintronic computational RAM (CRAM) based on STT-MRAM has demonstrated superior compute performance across various workloads, including neural networks [5]. At the component level, logic and bit-wise arithmetic operations in MRAM are performed based on iterative read/write operations. For practical applications, cell-level error rates originating from the probabilistic magnetization switching must be carefully controlled to ensure functional correctness. We have recently demonstrated that error-aware fine-tuning and CMOS+MRAM heterogeneous design of the arithmetic components can achieve area-efficient and robust digital in-memory computation with MRAM.

Analog In-Memory and Neuro-Mimetic Computing Crossbar-based analog in-memory computing further enhances the processing parallelism of matrix-vector multiplication (MVM) computation by exploiting physics (Ohm's Law) to achieve analog-domain weighted sums. Analog crossbar mapping of synaptic weights is a natural platform for neuro-mimetic computing. Spintronics provides the underlying devices for neuro-synaptic components in crossbars. A binary probabilistically switched device can serve as a stochastic neuron [6], while domain wall motions and multi-grain switching can emulate analog accumulation of membrane potential in a leaky integrate and fire neuron [7], [8]. As for storing synaptic weight matrices, standard 1-bit MRAM needs bit-slicing and peripheral shift-and-add operations to process multi-bit workloads [9]. Spin-orbit-torque magnetic tunnel junctions (SOT-MTJs) with separate read and write paths offer a viable solution to mitigate circuit non-idealities by tuning the device conductance without write constraint [10]. Moreover, exploiting the granular nanostructure developed for hard drive media enables multi-bit MRAM-based synapses [11].

The array-level analog weighted sums from the crossbar computing cores need to be post-processed to obtain the software-level MVM results. Processing the array-level partial sums is typically done by the peripheral analog-digital converters (ADCs). Multi-bit ADCs are the area and energy bottlenecks in analog compute-in-memory systems. To mitigate such ADC bottlenecks, stochastic SOT-MTJs are exploited to convert analog-domain currents into stochastic bits with high area efficiency and low latency. It is observed that quantization and stochasticity-aware training can help achieve satisfiable DNN accuracy with only 2-4 samplings per conversion [12].

IV. CONCLUSION AND FUTURE DIRECTIONS

The continuous progress of fundamental spintronics and the advances of MRAM in the microelectronic ecosystem open up exciting opportunities for spintronics in AI hardware. Particularly, as the scale and complexity of AI workloads grow with the proliferation of large language models (LLMs), memory and storage are entering the center stage in the design of AI hardware. The evolving landscape of AI computation further opens up uncharted territories for spintronic hardware. For example, long-context LLM inference may potentially benefit from near-data processing and pre-processing, while mixture of experts (MoE) requires more sophisticated designs of interconnects among a large number of computing cores. The area efficiency of MRAM, combined with the rich physics of spintronics from multi-domain dynamics to stochastic switching, create a exciting venue for device-to-algorithm co-exploration.

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